POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Electrical Engineering Materials | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by wave function? Derive the wave function for an electron confined in the infinite potential well. 8

   b) Define density of state function in quantum mechanics. Show that the density of state varies with square root of energy with necessary plot. 7

2. a) Calculate mean scattering time and conductivity of conduction electrons in copper at room temperature, given that drift mobility of conduction electron is 43 $cm^2 V^{-1} S^{-1}$, density of copper is 8.96 g cm^{-3} and its atomic mass is 63.5 g/mol. 8

   b) What do you mean by electrical breakdown in gaseous medium? Explain with Townsend discharge mechanism. 7

3. a) What is polarization? Derive the Clausius –Mossetti equation for ionic polarization. 8

   b) Why hard magnetic material is preferred for making permanent magnet while soft magnetic material is used for high frequency application. Explain with hysteresis loop. 7

4. a) What is Fermi level? Prove that the position of Fermi level is near the middle of band gap in pure silicon semiconductor. 8

   b) Find the diffusion coefficients of electrons and holes of a silicon single crystal at 27^0 C if the mobilities of electrons and holes are 0.17 $m^2 V^{-1} S^{-1}$ and 0.025 $m^2 V^{-1} S^{-1}$ respectively at 27^0 C. 7

5. a) A glass dielectric has dielectric constant 2.6 and loss tangent 4×10^{-5} at 1MHz. Calculate the loss of power per unit volume if the signal applied to the glass has peak amplitude of 0.71 V/m. 7

b) A heavily doped p-side with acceptor concentration of 10^{18} cm^{-3} is connected to n-side with donor concentration of 10^{16} cm^{-3}. Calculate the built in potential and depletion width. Assume T = 300 K and n_i = 1.5 $\times 10^{10} cm^{-3}$.

6. a) Why is silicon favored over Germanium for semiconducting materials? With the use of a diagram, explain the crystal growth in the Floating Zone process.

b) Define relative permeability and susceptibility. Show that $M = H(\mu_r - 1)$, where symbols have their usual meaning.

c) Calculate the permeability and susceptibility of an iron bar of cross section area 0.2 cm^2 when a magnetising field of 1200 Am^{-1} produces magnetic field of 24 micro Weber.

7. Write short notes on: **(Any two)**

a) De-Broglie hypothesis

b) Kroning Penney model

c) Continuing Equation and its application

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POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2016
Programme: BE Full Marks: 100
Course: Electrical Engineering Materials Pass Marks: 45
Time : 3hrs.

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Attempt all the questions.

a) Derive time independent one dimensional Schrödinger Wave Equation .A particle is trapped in potential well with V(x)=0 , except at the boundaries x=0 and x=L where it is infinite large . Obtain solution for time independent one dimensional Schrödinger wave equation. 8

b) An electron is confined to a 1 micron thin layer of silicon. Assuming that the semiconductor can be adequately described by a one-dimensional quantum well with infinite walls, calculate the lowest possible energy within the material in units of electron volt. If the energy is interpreted as the kinetic energy of the electron, what is the corresponding electron velocity? (The effective mass of electrons in silicon is 0.26 m_0, where $m_0 = 9.11 \times 10^{-31}$ kg is the free electron rest mass). 7

a) A uniform silver wire has resistivity of 5×10^{-8} ohm-m at room temperature. For the electrical field along the wire of $1\,Vcm^{-1}$, compute average drift velocity of electron assuming there are 5.8×10^{28} conduction electron per m^3. Also calculate mobility and relaxation time of electron. 7

b) Derive the relation $m^* = \frac{h^2}{d^2E/dK^2}$ 8

a) A electrolyte condenser consisting of a oxidizing aluminum sheet with effective surface area $400 cm^2$ has capacitance of 8 μF .Relative dielectric constant of Al_2O_3 (ε_r) is 8. A potential difference of 10 V is applied between aluminum and electrode. What is field strength and dipole moment induced in oxide layer? 7

b) Classify the types of polarization and explain orientational and 8 interfacial polarization.

4. a) Prove that in thermal equilibrium for a given semiconductor, $n_o p_o$ product is a constant that depends only on temperature.(Laws of Mass Action)

b) Derive the continuity equation. The steady state excess hole concentration of $\Delta p = 10^{16}$ at (x=0) is injected to semi –infinite Silicon bar of cross section area $A = 10^{-3} cm^2$. If the hole diffusion length $L_p = 10^{-3}$ cm and hole life time $\tau_p = 10^{-6}$S then calculate steady state stored charge Q_p and hole current Ip (x=0).

5. a) Derive the expression for the contact potential with necessary assumption.

b) Aluminum is alloyed in to an n-type silicon sample ($N_d = 10^{16}$ cm^{-3} forming an abrupt junction of circular cross section, with diameter 0.052 cm. Assume that the acceptor concentration in the alloyed re-grown region is $N_a = 4 \times 10^{18}$ cm^{-3} .Calculate V_0, x_{n0}, x_{po}, Q_+, and maximum field and Sketch the charge density and E(x).

6. a) Classify the magnetic materials and explain the properties of magnetic materials.

b) Explain ion implantation process for the fabrication.

7. Write short notes on: **(Any two)**

a) Physical Interpretation of wave function

b) Fermi -Dirac distribution function

c) Electrical Conduction in liquid.

POKHARA UNIVERSITY

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Attempt all the questions.

1. a) Define Wave function. Starting with general solution of Schrodinger equation for 1-D case, as $\Psi(x) = A\sin\alpha x + B\cos\alpha x$, where the notations have their usual meanings, derive the expression for total energy and normalized wave function. 8

   b) Prove that the expression of fermi energy at absolute zero for free electron model is $E_f = 3.65 \times 10^{-19} n^{2/3}$(in ev), where n be the number of electrons per unit volume. 7

2. a) Electron in undoped gallium arsenide have a mobility of 8.8 cm^2/V.S. Calculate average time between collision. Calculate distance travel between two collision. The average velocity of electron is 10^7 m/sec. 7

   b) Explain how electrical conduction takes place in liquid? 8

3. a) Obtain Clausius- Massotti equation for electronic polarization. 8

   b) What is interfacial polarization? 3

   c) Calculate the permeability of an iron bar of cross-sectional area $0.2cm^2$ when a magnetizing field of $1200Am^{-1}$ produces magnetic field of 24 *microWeber*. 4

4. a) Distinguish between ferromagnetic, ferromagnetic and antiferrimagnetic materials. Give an example for each class of material. Discuss the various uses of ferrites. 8

   b) Explain the differences between Hard and soft magnetic materials. 4

   c) Distinguish between intrinsic and extrinsic semiconductors. 3

5. a) A p-n junction is formed at 300K. The accepter and donor concentration in p-side and n- side are 10^{18} cm^{-3} respectively. Calculate: 7

      i. Built in potential

ii. Width of depletion layer

iii. Maximum value of electric field.

(Given: $\eta_i = 1.45\times10^{10}$ cm-3 $\varepsilon = \varepsilon_0\, \varepsilon_r = (11.9) \times (8.85\times10^{-12}$ F/m)

b) Explain the term depletion layer across a PN junction. Obtain the expression for the width of the depletion layer in terms of the impurity concentration and barrier potential.

6. a) Write Short note on Czochralski method.

b) "When donor type impurities are added to a semiconductor the concentration of holes decreases" Explain with reasons.

c) Explain the ion implantation process with a neat sketch.

7. Write short notes on: **(Any two)**

a) Heisenberg's uncertainty principle

b) Dielectric constant

c) Photolithography

POKHARA UNIVERSITY

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Attempt all the questions.

a) Define density of states function. Show that it varies parabolically with energy. 9

b) Calculate the wavelength of a proton travelling at 1900 ms^{-1}. 6

2. a) Obtain expression to evaluate effective mass of electron. Also interpret negative effective mass. 8

   b) Drift mobility of conduction electrons is $43cm^2V^{-1}S^{-1}$ and mean speed is $1.2 \times 10^6 m/s$. Calculate the mean free path of electrons between collisions. 7

3. a) Explain how electrical conduction takes place in gases? 7

   b) Explain different polarization mechanisms in materials. 8

4. a) Distinguish between paramagnetic and diamagnetic materials. Give an example for each class of materials. 5

   b) Distinguish between soft and hard magnetic materials.

   c) Calculate the field intensity of a magnetic field and the intensity of magnetization if 0.2A current is passed through a winding of 20turns/cm over an iron anchor ring having magnetic field 1.26T.

5. a) Show that Fermi level of intrinsic semiconductor lies midway between valence band and conduction band.

   b) The intrinsic resistivity of germanium at 300k is 0.45 Ωm. The electron mobility at 300K is 0.39 m^2/v-s. The hole mobility at 300K in Germanium is 0.19 m^2/v-s. Calculate the density of electrons in the intrinsic material. Also calculate the drift velocity of holes and electrons for an electric field E= 10^4 V/m.

6. a) Explain with necessary derivation about the Einstein's relationship between mobility and diffusion coefficient.

b) What are the advantages of Silicon over Germanium? Write t
advantages of ion implantation process over diffusion process.

7. Write short notes on: **(Any two)**

a) Compensation doping

b) Dielectric breakdown

c) Czochralski method of crystal growth

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Attempt all the questions.

1. a) Explain the Energy well model of metal and Show that the stationary energy levels of the particle are given by $E_n = \frac{n^2 h^2}{8mL^2}$. 8

   b) If longest wavelength required photoelectric effect for a certain metal is $2250\overset{0}{A}$. 7
      i. Find Work function of the metal
      ii. The velocity of electrons ejected from the metal surface if the same metal is radiated with $\lambda = 1800\overset{0}{A}$.
      iii. Kinetic energy of ejected electrons.

2. a) State Fick's law of diffusion. Derive Einstein's relationship between mobility and diffusion coefficient. 8

   b) Explain how electrical conduction takes place in liquid. 7

3. a) What is polarization? Show that "there will be no polarization if ε_r=1". 8

   b) Define local electric field. With the help of this obtain the expression for Clausius-Massotti equation. 7

4. a) Explain the domain structure and domain wall of magnetic material. 7

   b) What is magnetization, explain the types of magnetization? 8

5. a) What is minority carrier suppression? Prove electron concentration and conduction in n-type semiconductor is defined by donor impurity. 9

   b) A Si sample has been doped with 10^{17} arsenic atoms cm^{-3}. Calculate the conductivity of sample at 27 °C and at 127 °C. Given, μ_e= 800 $cm^2 V^{-1}S^{-1}$ at 27 °C, μ_e= 420 $cm^2 V^{-1}S^{-1}$ at 127 °C and n_i= 1.45×10^{10} cm^{-3}. 6

6. a) A n- type semiconductor containing 10^{16} Phosphorus (donor) atoms cm^{-3} has been doped with 10^{17} Boron(acceptor) atoms cm^{-3}. Calculate electron and hole concentration in this semiconductor ($n_i = 1.45 \times 10^{10} cm^{-3}$).

b) Explain ion-implementation process. What are its advantages and disadvantages over diffusion process?

7. Write short notes on: **(Any two)**

a) Seebeck effect

b) Soft and Hard magnetic materials

c) Built in potential

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Year : 2018
Programme: BE
Full Marks: 100
Course: Electrical Engineering Materials
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by wave function? Derive the time independent form of Schrodinger's wave equation. 8

   b) Find the temperature at which there is 98% probability that a state 0.3eV below the Fermi energy level will be occupied by an electron. 7

2. a) Define the term electrons drift velocity and mobility. How are they related? Show that the conductivity of electrons with in metallic conductor is product of charge density and mobility. 9

   b) Molybdenum has the BCC crystal structure, a density of 10.22 g cm^{-3}, and an atomic mass of 95.94 g mol^{-1}. What is atomic concentration, lattice parameter and atomic radius of molybdenum? 6

3. a) Show that ionic conductivity depends inversely with temperature. 7

   b) Write the types of polarization. Derive the Clausius-Massotti equation for electronic polarization, relating to polarizibility with the permittivity. 8

4. a) Explain then eddy current loss and hysteresis loss in magnetic materials? 8

   b) Calculate the permeability and susceptibility of an iron bar of cross-sectional area 0.2 cm^2 when a magnetic field strength of 1200 A/m produces magnetic flux of 24 micro weber. 7

5. a) Show that the Fermi level of intrinsic semiconductor lies midway between the valance band and the conduction band. [$E_F = E_G/2$]. 8

   b) A silicon sample is doped with 10^{15} Antimony atoms/cm^3. Find carrier concentration, its resistance and the shift in Fermi level from its intrinsic Fermi level at 27°C. If it is further doped with 2×10^{17} boron atoms/cm^3, what is the new resistance?